

Forecasting the economic impact of AI - structural tasks ->role aggregation

BACKGROUND

In July 2025, Faculty's CEO (now Accenture CTO), Marc Warner, convened a team to investigate the economic impact of AI's rapidly increasing capabilities. That programme is organised as a three-stage pipeline: **constraints/technologies -> tasks -> jobs/roles -> economy**.

This workstream targets the **Stage 1 -> Stage 2 seam**: how task-level automation forecasts are turned into statements about *jobs/roles*. Previous work (task->role mapping via O*NET & Blue Rose) has been relying on a **linear, time-weighted roll-up** $\text{occupation_index} = \sum_k w_k * p_k$. This is an accounting identity that ignores non-separability/O-ring effects, endogenous task weights, reinstatement/new tasks, remaining-task expertise, and the gap between technical feasibility and economic profitability.

Contribution

Overarching objective: **replace the naive linear roll-up with an economically-grounded job-production model** that turns task-level automation *forecasts* into occupation-level automation forecasts, with uncertainty. This contributes to the overall "**constraints -> tasks -> jobs -> labour market**" framework, while improving on current aggregation methodology.

The contribution is **not** assessed against the academic economics novelty bar, but against the **AI-automation-forecasting field** - OpenAI papers, Anthropic Economic Index, AISI/IFOW - which still uses the linear time-weighted roll-up. We believe contributions can be twofold:

- Bringing the best current economic structure into a forecasting field that still uses the accounting identity.
- Showing that it changes the answers on labour market impacts: different occupation rankings, different displacement timing, different sector winners/losers, etc.

Applied research from main labs

- **OpenAI, "The AI Jobs Transition Framework" (2026)** ([note](#)) - maps near-term AI pressure across 921 occupations by bolting human-necessity + demand-elasticity filters onto the linear task-exposure roll-up; does not fix within-occupation aggregation.
- **Anthropic Economic Index (2026)** ([pdf](#)) - measures real Claude usage mapped to O*NET tasks/occupations (automation vs. augmentation), still aggregating task coverage to occupations by time-share.
- **Eloundou et al., "GPTs are GPTs" (2023)** ([pdf](#)) - the foundational task-level LLM exposure measure (share of occupation task time exposed to LLMs) that the lab frameworks build on.

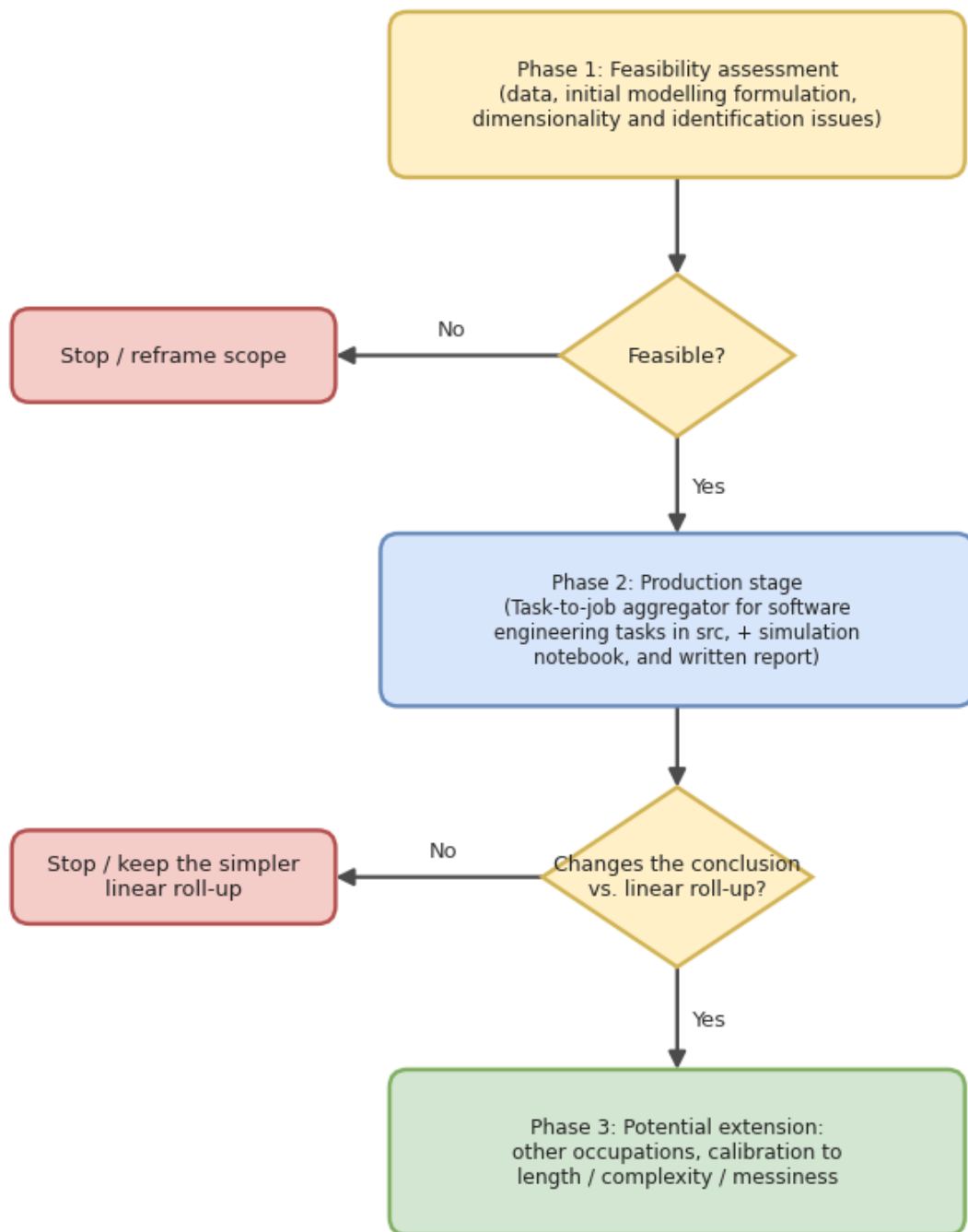
Economic literature

- **Gans & Goldfarb, "O-Ring Automation" (2026)** ([note](#)) - theoretical microfoundation: proves linear exposure indices overstate displacement under complementarity, adoption is threshold/bundled, and labour income can rise under partial automation.
- **Hampole, Papanikolaou, Schmidt & Seegmiller (2025, NBER WP 33509)** ([note](#)) - closest structural work: CES over tasks with task-specific capital substituting for labour + endogenous time reallocation; two sufficient statistics (mean exposure, concentration), causally estimated via a hiring-network IV.

- **Autor & Thompson, "Expertise" (2025)** ([note](#)) - the expertise of the *remaining* tasks drives the wage/employment direction after automation (a wage/skill layer, not an aggregator).
- **Acemoglu, "The Simple Macroeconomics of AI" (2025)** ([note](#)) - only the *profitable* share of exposed tasks automates -> modest macro bound (~0.66% TFP / ~1% GDP over 10 yrs).
- **Svanberg et al., "Beyond AI Exposure" (2024)** ([note](#)) - empirical feasibility-vs-profitability gate: only ~23% of vision-task wages are currently cost-effective to automate.

Scope and deliverables

The project runs as a phased tree gated at two points: a **feasibility gate** before the production stage, and a **decision-relevance gate** after it (does the structural aggregator change the conclusion vs. the linear roll-up?). Only if both pass do we reach the **potential extension** beyond software engineering tasks.



Phase 1 - Feasibility assessment (first deliverable)

Can the structural aggregator be specified, calibrated, and validated with the data available? Assess:

- **Data:** task-level p (Luo et al. 2026, software), O*NET task statements + DWA/GWA hierarchy + importance/frequency weights, Hampole et al. (2025) estimates for calibration, and AI-rollout shocks for validation (TBC).
- **Initial modeling formulation:** the production technology (Nested CES within tasks + Leontief bottleneck across them; endogenous time allocation; profitability gate)

- **Dimensionality and Identification:** how many free parameters/elasticities it requires; can human-AI and task-task substitution be identified from available data, or only borrowed from Hampole's IV estimates as calibration? -> a **calibrate-vs-estimate** call.
- **Uncertainty:** can noisy Stage-1 p (small-sample/judge variance) be propagated through the aggregator tractably (analytic vs. Monte Carlo)?

Gate 1 - feasible?

- **No** -> stop or reframe the scope.
- **Yes** -> unlock the production stage below.

Phase 2 - Production stage: Tasks-to-job aggregator

A structural **job-production aggregator** (CES/nested + Leontief bottleneck) that replaces the linear roll-up, calibrated for one software occupation using Luo et al. (2026) estimates.

- **Input:** task-level automation probabilities/forecasts from workstream 1 (task automation forecasts; Luo et al. 2026) and O*NET time weights.
- **Output:** occupation-level automation timing/curve with propagated uncertainty, compared against the naive linear-sum baseline.
- **Deliverable:** reference implementation in `src/` + a runnable `simulations/` notebook + a report (motivation, methodology, results, comparison to linear roll-up).

Gate 2 - does it change the conclusion?

- **No** (the structural aggregator reproduces the linear roll-up's occupation rankings/timing) -> keep the simpler linear roll-up.
- **Yes** (materially different rankings and/or displacement timing) -> extend to other occupations.

Phase 3 - Potential extension

- Other **occupations and industries** (GDPval / EnterpriseVal as task coverage broadens and data access lands).
- **Calibration to task length l , complexity and messiness m** across the broader task set.

Timeline and resourcing

A single view of phases, effort, and estimated deliverable dates under two resourcing scenarios.

- **Scenario 1:** Esther 1 day/week in Phase 1; a 50% data scientist (2.5 d/wk) added in Phase 2 -> Phase 2 capacity ~3.5 d/wk.
- **Scenario 2:** Esther 2 days/week in Phase 1; a 50% data scientist (2.5 d/wk) added in Phase 2 -> Phase 2 capacity ~4.5 d/wk.
- Dates are approximate ("week of"), assume steady weekly cadence and no further holiday gaps; the two gates are decision points.

Phase / task	Est. work-days	Scenario 1 (Esther 1 d/wk)	Scenario 2 (Esther 2 d/wk)
Phase 1 - Feasibility assessment	~12 d	week of Oct 26	week of Sep 14
- Foundation + literature review	~2 d	week of Aug 17	week of Aug 10
- Feasibility: data	~2 d	week of Aug 31	week of Aug 17
- Feasibility: initial formulation	~3 d	week of Sep 21	week of Aug 31

- Feasibility: dimensionality and identification	~3 d	week of Oct 12	week of Sep 7
- Feasibility: uncertainty propagation	~2 d	week of Oct 26	week of Sep 14
- Gate 1: feasible? (go/no-go + scoping note)	-	~Oct 30	~Sep 18
+ data scientist @ 2.5 d/wk added from Phase 2		capacity 3.5 d/wk	capacity 4.5 d/wk
Phase 2 - Production stage: tasks-to-job aggregator	~34 d	week of Jan 4, 2027	week of Nov 9
- Specification (CES + Leontief bottleneck; profitability gate)	~10 d	week of Nov 16	week of Oct 5
- Simulations using Luo (2026) estimates on software	~6 d	week of Nov 30	week of Oct 12
- Calibration / estimation + uncertainty propagation	~6 d	week of Dec 14	week of Oct 19
- Validation vs. linear roll-up	~6 d	week of Dec 21	week of Nov 2
- Gate 2: changes the conclusion?	-	~late Dec	~Nov 6
- Brief / report writing	~6 d	week of Jan 4, 2027	week of Nov 9
Phase 3 - Potential extension	TBD	-	-
- Extend to other occupations & industries (GDPval / EnterpriseVal)	TBD	-	-
- Calibration to task length ℓ , complexity/messiness m	TBD	-	-